



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: GRID-INDIA/NLDC/SO/Daily PSP Report

दिनांक: 21.11.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 20.11.2024.

महोदय/Dear Sir,

आईईजीसी-2023 की धारा स.-38(1) के प्रावधान के अनुसार, दिनांक 20-नवंबर-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 38(1) of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 20.11.2024, is available at the NLDC website.

धन्यवाद,
Thanks

Report for previous day

Date of Reporting: 21-Nov-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 19:00 hrs; from RLDCs)	51940	58928	43712	21256	2633	178469
Peak Shortage (MW)	0	0	0	0	0	0
Energy Met (MU)	1146	1437	1044	435	49	4112
Hydro Gen (MU)	113	32	91	28	17	281
Wind Gen (MU)	5	59	68	-	-	132
Solar Gen (MU)*	148.72	90.38	89.18	2.89	1.05	332
Energy Shortage (MU)	0.21	0.00	0.00	0.03	0.00	0.24
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	57693	67675	50198	22695	2904	196812
Time Of Maximum Demand Met	10:29	10:09	09:27	17:46	17:34	09:41

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.022	0.00	0.00	1.23	1.23	84.09	14.69

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	7603	0	144.0	64.2	-0.2	766	0.00
	Haryana	7415	0	144.3	102.7	-1.3	235	0.00
	Rajasthan	16845	0	320.7	115.5	-0.7	465	0.00
	Delhi	3890	0	73.0	67.6	1.3	307	0.00
	UP	17180	0	321.6	119.0	-1.0	514	0.00
	Uttarakhand	2167	0	41.2	32.6	-0.1	101	0.00
	HP	2056	0	36.3	27.4	-0.6	137	0.00
	J&K(UT) & Ladakh(UT)	2751	100	56.7	49.5	0.5	168	0.21
	Chandigarh	205	0	3.6	3.5	0.1	71	0.00
WR	Railways_NR ISTS	212	0	4.3	3.9	0.4	45	0.00
	Chhattisgarh	4301	0	87.4	41.2	-0.5	284	0.00
	Gujarat	20622	0	412.6	194.1	0.4	499	0.00
	MP	17309	0	330.3	204.8	-2.5	650	0.00
	Maharashtra	23873	0	531.1	184.7	-0.7	602	0.00
	Goa	678	0	14.4	13.7	0.1	28	0.00
	DNHDDPDCL	1323	0	30.4	30.4	0.0	55	0.00
	AMNSIL	855	0	18.3	9.5	-0.1	265	0.00
	BALCO	531	0	12.7	12.7	0.0	10	0.00
SR	Andhra Pradesh	10112	0	201.2	47.3	-1.0	486	0.00
	Telangana	10302	0	197.9	102.0	0.2	373	0.00
	Karnataka	13039	0	235.6	71.6	1.2	749	0.00
	Kerala	4283	0	85.6	68.8	1.1	529	0.00
	Tamil Nadu	15607	0	315.0	190.8	-0.6	591	0.00
	Puducherry	410	0	9.1	8.9	-0.4	37	0.00
ER	Bihar	4519	0	85.4	72.3	2.2	317	0.03
	DVC	3073	0	64.6	-42.9	-0.5	302	0.00
	Jharkhand	1903	0	36.0	24.1	-1.2	429	0.00
	Odisha	5643	0	105.9	31.1	-1.1	561	0.00
	West Bengal	7351	0	141.1	13.2	-2.0	216	0.00
	Sikkim	105	0	1.7	1.6	0.1	29	0.00
	Railways_ER ISTS	12	0	0.1	0.2	0.0	0	0.00
NER	Arunachal Pradesh	155	0	2.8	2.6	0.0	25	0.00
	Assam	1697	0	29.0	21.4	0.5	124	0.00
	Manipur	200	0	2.5	2.5	0.0	40	0.00
	Meghalaya	367	0	6.3	5.4	-0.2	70	0.00
	Mizoram	139	0	2.2	1.8	0.0	27	0.00
	Nagaland	154	0	2.4	2.2	0.0	27	0.00
	Tripura	252	0	4.1	4.8	0.1	46	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	1.0	3.8	-23.2	-16.4
Day Peak (MW)	95.5	392.4	-982.0	-762.0

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	230.5	-154.1	120.9	-200.3	3.0	0.0
Actual(MU)	217.1	-152.7	121.2	-198.0	3.4	-9.1
O/D/U/D(MU)	-13.4	1.3	0.3	2.3	0.4	-9.1

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	4532	11746	6008	3821	655	26761	48
State Sector	9568	9429	8281	1459	339	29076	52
Total	14100	21175	14289	5280	994	55838	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	654	1446	591	679	9	3378	76
Lignite	26	14	48	0	0	88	2
Hydro	113	32	91	28	17	281	6
Nuclear	36	51	76	0	0	163	4
Gas, Naptha & Diesel	6	13	6	0	24	49	1
RES (Wind, Solar, Biomass & Others)	174	151	180	5	1	512	11
Total	1009	1708	992	712	51	4471	100
Share of RES in total generation (%)	17.23	8.84	18.16	0.75	2.12	11.44	
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	31.98	13.75	35.03	4.62	35.39	21.38	

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.022
Based on State Max Demands	1.062

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	196812	9:41	0
Non-Solar hr	192492	18:20	76

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 06:00 to 18:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES								
Import=(+ve) /Export =(-ve) for NET (MU) Date of Reporting: 21-Nov-2024								
Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	303	0.0	7.4	-7.4
2	HVDC	PUSAULI B/B	-	2	299	0.0	6.7	-6.7
3	765 kV	GAYA-VARANASI	2	0	1080	0.0	17.1	-17.1
4	765 kV	SASARAM-FATEHPUR	1	20	484	0.0	6.2	-6.2
5	765 kV	GAYA-BALIA	1	0	712	0.0	13.1	-13.1
6	400 kV	PUSAULI-VARANASI	1	0	166	0.0	3.5	-3.5
7	400 kV	PUSAULI -ALLAHABAD	1	0	173	0.0	3.0	-3.0
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	16	520	0.0	7.2	-7.2
9	400 kV	PATNA-BALIA	2	0	762	0.0	12.6	-12.6
10	400 kV	NAUBATPUR-BALIA	2	0	359	0.0	3.5	-3.5
11	400 kV	BIHARSHARIFF-BALIA	2	97	144	0.0	1.5	-1.5
12	400 kV	MOTIHARI-GORAKHPUR	2	0	456	0.0	7.8	-7.8
13	400 kV	BIHARSHARIFF-VARANASI	2	102	391	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	0	118	0.0	0.3	-0.3
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.0	0.0	0.0
16	132 kV	GARWAH-RIHAND	1	30	0	0.5	0.0	0.5
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						0.5	89.9	-89.4
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	487	335	0.3	0.0	0.3
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	448	594	0.0	1.6	-1.6
3	765 kV	JHARSUGUDA-DURG	2	0	1200	0.0	23.2	-23.2
4	400 kV	JHARSUGUDA-RAIGARH	4	0	661	0.0	10.2	-10.2
5	400 kV	RANCHI-SIPAT	2	33	302	0.0	3.1	-3.1
6	220 kV	BUDHIPADAR-RAIGARH	1	0	85	0.0	1.2	-1.2
7	220 kV	BUDHIPADAR-KORBA	2	41	116	0.0	0.8	-0.8
ER-WR						0.3	40.0	-39.8
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	206	0.0	4.9	-4.9
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	1196	0.0	29.0	-29.0
3	765 kV	ANGUL-SRIKAKULAM	2	0	2780	0.0	57.2	-57.2
4	400 kV	TALCHER-I/C	2	644	0	12.8	0.0	12.8
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	91.0	-91.0
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	0	449	0.0	8.0	-8.0
2	400 kV	ALIPURDUAR-BONGAIGAON	2	0	645	0.0	9.8	-9.8
3	220 kV	ALIPURDUAR-SALAKATI	2	0	105	0.0	1.1	-1.1
ER-NER						0.0	18.9	-18.9
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	0	666	0.0	15.7	-15.7
NER-NR						0.0	15.7	-15.7
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	512	0.0	12.0	-12.0
2	HVDC	VINDHYACHAL B/B	-	297	0	5.7	0.0	5.7
3	HVDC	MUNDRA-MOHINDERGARH	2	0	1905	0.0	30.3	-30.3
4	765 kV	GWALIOR-AGRA	2	0	1584	0.0	24.2	-24.2
5	765 kV	GWALIOR-PHAGI	2	416	2032	0.8	27.5	-26.7
6	765 kV	JABALPUR-ORAI	2	0	904	0.0	30.4	-30.4
7	765 kV	GWALIOR-ORAI	1	953	0	14.3	0.0	14.3
8	765 kV	SATNA-ORAI	1	0	1046	0.0	20.5	-20.5
9	765 kV	BANASKANTHA-CHITORGARH	2	1432	0	15.9	0.0	15.9
10	765 kV	VINDHYACHAL-VARANASI	2	0	2116	0.0	37.5	-37.5
11	400 kV	ZERDA-KANKROLI	1	291	0	3.0	0.0	3.0
12	400 kV	ZERDA -BHINMAL	1	202	0	2.1	0.0	2.1
13	400 kV	VINDHYACHAL -RIHAND	1	973	0	21.2	0.0	21.2
14	400 kV	RAPP-SHUJALPUR	2	603	221	3.1	1.4	1.7
15	400 kV	NEEMUCH-Chittorgarh	2	652	261	3.4	1.5	1.9
16	220 kV	BHANPURA-RANPUR	1	0	193	0.0	3.5	-3.5
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.3	-2.3
18	220 kV	MEHGAON-AURAIYA	1	142	0	1.7	0.0	1.7
19	220 kV	MALANPUR-AURAIYA	1	105	0	0.9	0.0	0.9
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						72.1	191.0	-118.9
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	307	0.0	7.2	-7.2
2	HVDC	RAIGARH-PUGALUR	2	0	1002	0.0	15.6	-15.6
3	765 kV	SOLAPUR-RAICHUR	2	231	1057	0.1	7.9	-7.8
4	765 kV	WARDHA-NIZAMABAD	2	0	1554	0.0	27.6	-27.6
5	765 kV	WARORA-WARANGAL(NEW)	2	0	2051	0.0	36.9	-36.9
6	400 kV	KOLHAPUR-KUDGI	2	1075	0	19.5	0.0	19.5
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	0	0.0	0.0	0.0
9	220 kV	XELDEM-AMBEWADI	1	0	122	2.2	0.0	2.2
WR-SR						21.8	95.3	-73.5
INTERNATIONAL EXCHANGES								
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Import(+ve)/Export(-ve) Energy Exchange (MU)		
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	115	68	94	2.25		
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	163	35	115	2.75		
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	-211	-163	-189	-4.54		
	NER	132kV GELEPHU-SALAKATI	16	1	6	0.14		
	NER	132kV MOTANGA-RANGIA	28	4	16	0.39		
NEPAL	NR	NEPAL IMPORT (FROM UP)	0	0	0	0.00		
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	60	0	32	0.78		
	ER	NEPAL IMPORT (FROM BIHAR)	0	0	0	0.00		
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	332	22	126	3.03		
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-930	-918	-922	-22.12		
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-762	-402	-685	-16.44		
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-52	0	-45	-1.08		

CROSS BORDER EXCHANGE SCHEDULE

Date of Reporting: 21-Nov-2024

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	0.00	0.00	0.00	0.22	0.00	0.00	0.22
Nepal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bangladesh	23.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.17
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	23.17	0.00	0.00	0.00	0.00	0.22	0.00	0.00	23.39

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.89	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.89
Nepal	0.00	0.00	0.00	0.00	0.00	3.53	0.00	0.00	3.53
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	0.89	0.00	0.00	0.00	0.00	3.53	0.00	0.00	4.42

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.89	0.00	0.00	0.00	0.00	-0.22	0.00	0.00	0.67
Nepal	0.00	0.00	0.00	0.00	0.00	3.53	0.00	0.00	3.53
Bangladesh	-23.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-23.17
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	-22.28	0.00	0.00	0.00	0.00	3.31	0.00	0.00	-18.97

Date of Reporting: 21-Nov-2024												
15 Min (INSTANTANEOUS) ALL INDIA GRID FREQUENCY, GENERATION & DEMAND MET (SCADA DATA)												
TIME	FREQUENCY (Hz)	DEMAND MET (MW)	NUCLEAR (MW)	WIND (MW)	SOLAR (MW)	HYDRO** (MW)	GAS (MW)	THERMAL (MW)	OTHERS* (MW)	NET DEMAND MET (MW)	TOTAL GENERATION (MW)	NET TRANSNATIONAL EXCHANGE (MW) (+ve) Import, (-ve) Export
		(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I=A-(C+D))	(J=B+C+D+E+F+G+H)	(K)
0:00	50.08	148096	6454	7226	48	7979	1629	126118	1941	140822	151395	-526
0:15	49.98	147518	6447	7474	48	7589	1721	125516	1998	139996	150793	-528
0:30	50.03	147223	6430	7656	49	7602	1732	125026	2034	139518	150529	-497
0:45	49.98	146175	6430	7571	49	7325	1635	124355	1995	138555	149360	-499
1:00	49.98	145006	6444	7535	48	7258	1628	123457	2001	137423	148371	-515
1:15	49.98	144253	6437	7668	48	7064	1751	122649	1958	136537	147575	-473
1:30	49.98	143146	6448	7782	47	7037	1637	121573	1956	135317	146480	-472
1:45	50.03	143168	6452	7807	48	7227	1638	121238	2012	135313	146422	-492
2:00	50.03	141578	6437	7721	47	7234	1626	119861	1987	133810	144913	-509
2:15	49.98	141385	6438	7764	48	7151	1800	119707	1963	133573	144871	-484
2:30	50.03	140993	6444	7966	48	6774	1805	119364	2056	132979	144457	-486
2:45	50.04	140760	6445	7934	47	6708	1631	119109	1917	132779	143791	-482
3:00	50.03	140177	6446	7986	47	6174	1646	119120	1869	132144	143288	-490
3:15	49.98	140470	6442	8031	47	6174	1639	119454	1905	132392	143692	-488
3:30	49.98	140272	6445	8080	41	6069	1826	119404	1884	132151	143749	-502
3:45	49.98	140908	6445	8067	40	6128	1637	120006	1893	132801	144216	-557
4:00	50.03	140541	6435	8213	40	6196	1813	119560	1877	132288	144134	-570
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7:15	49.99	184326	6454	5626	2939	19680	1810	150433	1866	175761	188808	-920
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